

Name	Element type	Palette group	Property	Value
CreateQuestions	Screen	NA	Title	CreateQuestions
HorizontalArrangement1	HorizontalArrangement	Layout	Width	FillParent
Label1	Label	User interface	Text	'Questions Set:'
QpaperName	TextBox	User interface	Hint width	'Name of Question Paper'
				FillParent
HorizontalArrangement2	HorizontalArrangement	Layout	Width	FillParent
Label2	Label	User interface	Text	'Questions'
Question	TextBox	User Interface	Hint Width Multiline	'Type the Question' FillParent checked
HorizontalArrangement3	HorizontalArrangement	Layout	Width	FillParent
Label3	Label	User interface	Text	'Answer'
QPaperName	TextBox	User interface	Hint width	'Type the Answer' FillParent
HorizontalArrangement4	HorizontalArrangement	Layout	Width	Automatic
SaveQnA	Button	User interface	Text	Save Question
GoBack	Button	User interface	Text	'Done (Go Back)'
TinyDB1	TinyDB	Storage		

Table 4: CreateQuestions' screen elements

TinyDB has some very simple interfaces for CRUD (create, read, update and delete) operations. In fact, create and update are one and the same operation as the presence of a tag before them does not make any difference. In addition, a GetTags interface allows one to read all the tags present in the database. The StoreValue interface allows one to store a value against a particular tag. The GetValue interface allows the reading of the value stored against a particular tag. There is also a ClearTag interface, which deletes a tag and its value from the database. Putting these together we have the blocks for the CreateQuestions screen, as shown in Figure 5.

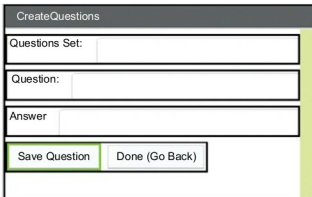


Figure 4: CreateQuestions UI



Figure 5: CreateQuestions blocks

AnswerQuestions screen

Now we come to the AnswerQuestions screen. The elements here are similar to CreateQuestions, except that we add an element for voice input, called SpeechRecognizer. Also, we create an image button to indicate voice input is enabled. To do this, we can upload the image of a microphone into the project's media section. In the current case, I have uploaded a file *microphone.png* into the project. Then we can set the image for the button. Next, let's add a couple of buttons to navigate forward and back, amongst the questions. We describe elements of the screen in Table 5. The screen itself is shown in Figure 6.



Note: There are three values for width, height, etc. These could be automatic, which means that they will get adjusted based on their properties, like a text box getting adjusted based on text input. It could also be 'Fill Parent', which means that it will fill as much as the parent element allows it to; e.g., for our 'Mike' button, we want its height to fill as much as the parent will allow. Automatic will make it too big, based on the image size. We can also make it a percentage of the available space and, in this case, you can adjust the percentage out of 100 per cent to different elements available.

The blocks for AnswerQuestions in Figure 7 are similar to those of CreateQuestions except for additional checks and controls over the following items.

Validating the answer: Once the answer is entered, we need to validate it against the answer stored for the question when it was created. While doing this, I decided to show an incorrect answer by changing the colour of the answer to red. When the correct answer is entered, it changes back to black.

SpeechRecognizer input: When the 'Mike' button